Project Proposal

Adapting Multi-Level Knowledge Distillation

for NLP text Classification tasks

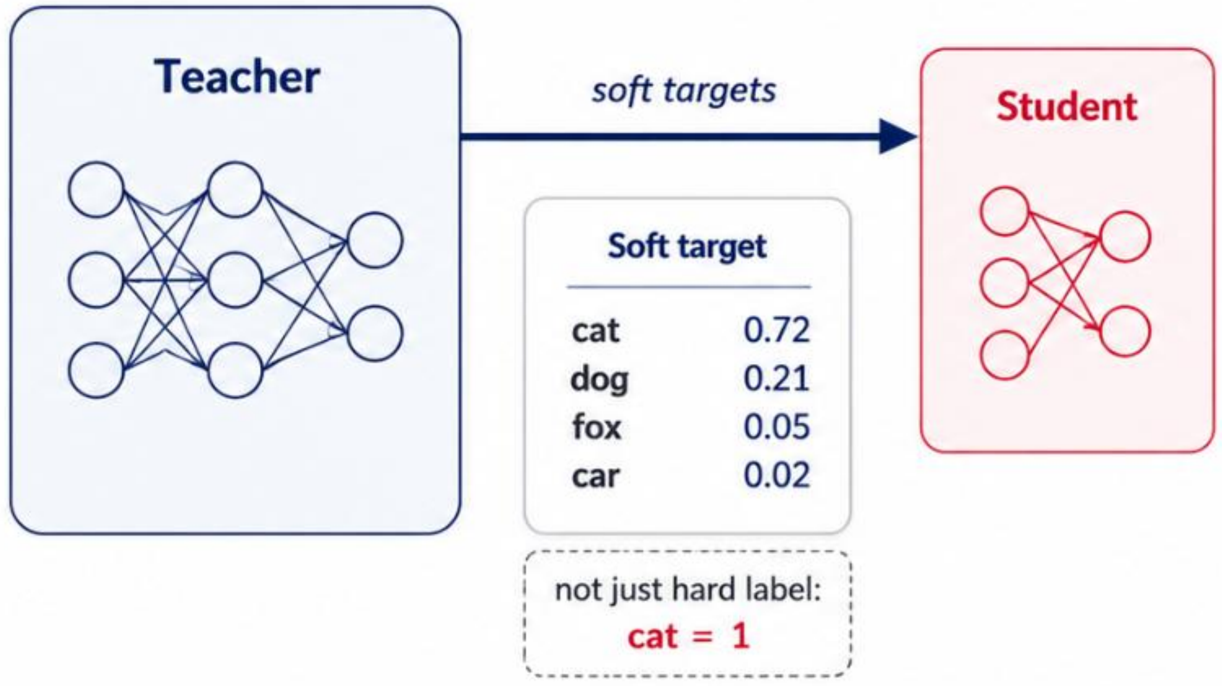
Per-level ablation of logit / hidden-state / attention distillation
for a broad spectrum of NLP classification tasks

Team 18
Spring 2026

Prior Work

Starting Point: Distilling the Knowledge in a Neural Network

NeurIPS Workshop, Hinton, Vinyals & Dean (2015)



- Teacher-Student learning with softened outputs
- Transfers class similarity information
  => Dark Knowledge
- The foundation of logit-level KD methods

A compact student learns the teacher's full class distribution, not just the correct class.

Prior Work

Compression is Now a First Class Concern

Large models turned compression into a deployment problem

Why It Matters Now

1. Models became expensive to deploy at scale
2. KD reduces size and inference cost while keeping quality
3. What to distill to preserve better quality?

Key Milestones

DistilBERT’19	40% smaller, 60% faster
TinyBERT’20	4-layer 14.5M params student
MiniLLM’24	KD for general LLMs
Targeted Distill’25	Task specialized distillation

*The live question is no longer **if** we distill, but **what** to distill.*

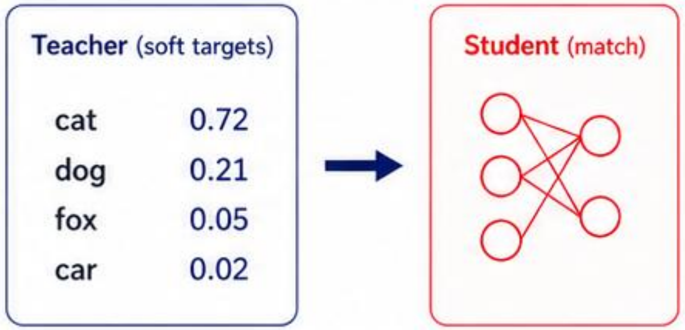
Prior Work

Three Signals a Student Can Learn From

KD can target the teacher's output, internal representations, or token relations

LOGIT

Output-distribution matching

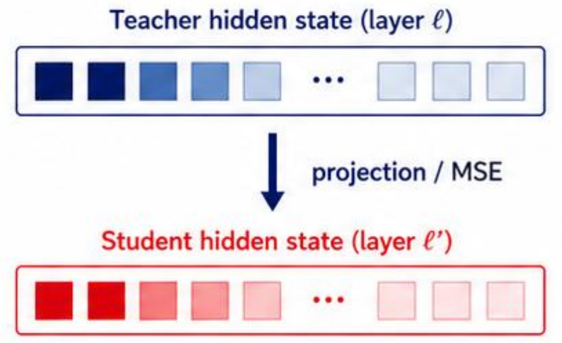


Learns the teacher's
<u>final class probabilities</u>

Cheap and task-agnostic,
but only final-layer knowledge

HIDDEN

Representation matching

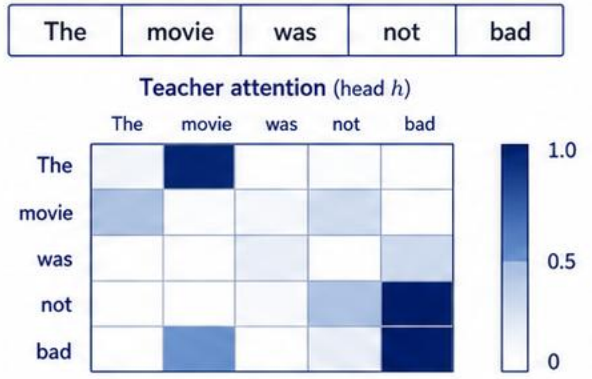


Learns the teacher's
<u>internal feature geometry</u>

Richer than logis,
but needs layer mapping/projection

ATTENTION

Relation-structure matching



Learns which tokens
<u>look at which</u>

Captures token relations and
is robust to architecture mismatch

Prior Work

TinyBERT — Unified Multi-Level Distillation

Jiao et al. · Findings of EMNLP 2020

TinyBERT: Distilling BERT for Natural Language Understanding

Four distillation levels

Embedding	*token-embedding MSE*
Hidden	*per-layer hidden-state MSE + projection*
Attention	*per-head attention-matrix MSE*
Logit	*soft-label KL on logits (Hinton-style)*

Design choices that matter for us

Two-stage training. General distillation on unlabeled text, then task-specific distillation on the target task.

Layer mapping. Student's 4 layers mapped to a subset of teacher's 12 — defines which hidden / attention tensors get matched.

Width projection. A learned W_h projects student hidden (312) into teacher hidden (768) before MSE.

Reported result. TinyBERT-4L recovers most of BERT-base's GLUE performance at ~14.5M parameters.

Prior Work

How to Distill Your BERT — Which Signal Actually Matters

Wang, Weissweiler, Schütze & Plank · ACL 2023 (Short)

How to Distill your BERT — an empirical study on weight initialisation & distillation objectives

What the paper asked

First systematic comparison of **intermediate-layer distillation (ILD)** objectives in both task-specific and task-agnostic settings.

Knobs studied

- ILD objective — vanilla KD / hidden-state MSE / attention KL / attention MSE.
- Student-layer initialisation — take which subset of teacher layers?
- Evaluated on GLUE tasks (SST-2, MNLI, QNLI, …).

Key findings

① Attention-transfer objectives give the best overall GLUE performance among ILD variants.

② Student-layer init matters a lot for vanilla KD and hidden-state KD — initialising from lower teacher layers is much better.

③ Attention KD is robust to initialisation choice.

Why this matters to us. Their experiments are GLUE-centric. Whether the same per-level ranking holds across **diverse text-classification tasks** (e.g., hate speech, NLI, sentiment, dialects) is an open question.

Prior Work
Limitations — What the Existing Work Does Not Answer

PAPER	CLAIM	GAP · WHAT WE ADDRESS
TinyBERT *Jiao et al., EMNLP 2020*	*Proposes multi-level KD — trains all levels simultaneously.*	→ Never quantifies the separate contribution of each level.
Beyond Logits *Gong et al., ACL 2025*	*Argues "logits alone are not enough."*	→ Never isolates which non-logit signal carries the improvement.
WID *Wu et al., NAACL 2024*	*Claims explicit alignment losses are unnecessary.*	→ So what value do hidden / attention KD actually add when kept?
How to Distill Your BERT *Wang et al., ACL 2023*	*Finds attention transfer best on GLUE (binary / 3-way tasks).*	→ Does the same ranking hold across diverse classification tasks?
Targeted Distill. / CompEffDist *Zhang et al., EMNLP 2025*	*Sentiment-specialized KD — but LLM teachers, generative students.*	→ Encoder classifier x multi-level x diverse classification tasks :still missing.
Cross-domain *(general observation)*	*Per-level ranking likely varies by dataset / domain.*	→ No head-to-head test across multiple task families and granularities.

COMMON THREAD No prior work runs a controlled per-level ablation on encoder-based text classification across this many task families — that intersection is ours.

Improvement Idea

Positioning: Fix the Framework, Ablate the Signal

POSITIONING STATEMENT

*We fix **TinyBERT's unified multi-level KD framework** and perform a **layer-wise impact / activation ablation** to re-examine — **per dataset** — which distillation signal matters most for **text classification** tasks — hate-speech, NLI, sentiment, and dialects.*

Fixed framework

Inherit TinyBERT's 4-layer student, teacher-layer mapping, and projection — so the only thing that varies across runs is which KD signals are active.

Per-signal ablation

Five conditions turn {logit, hidden, attention} on/off independently — each level's contribution becomes a directly measurable difference.

Per-dataset readout

Run the same five conditions across four task families — hate-speech, NLI, sentiment, and dialects — for a per-task readout.

Research Question

What Exactly Are We Asking?

RQ 1 · PRIMARY

*How much does each multi-level KD component — **logit, hidden state, attention** — differentially contribute to text-classification performance?*

RQ 2 · SECONDARY

*Does this pattern **vary across task families** — hate-speech, NLI, sentiment, and dialects?*

Target: map the per-level KD contribution, and test its cross-task robustness.

Method

From FitNets to Multi-Level KD — Our Positioning

FitNets	→	TinyBERT	→	Ours
CV · feature hints (2015)		*NLP · multi-level KD (2020)*		*per-level ablation across tasks*

What we actually do

- Fix TinyBERT's multi-level KD framework (embedding / hidden / attention / prediction).
- Adapt it to four text-classification tasks: hate-speech, NLI, sentiment, and dialects.
- Run 5 controlled ablations that successively add / remove KD levels.
- Quantify each level's marginal contribution and check task transfer.

Experiment Setup
Models & Datasets

Models

Role	Model	Layers	Hidden	Params
Teacher	bert-base-uncased	12	768	~110M
Student	TinyBERT_General_4L_312D	4	312	~14.5M

Teacher fine-tuned first per dataset

Student trained under five KD conditions

Experiment Setup

Models & Datasets

TASK 1

Hate Speech

Vidgen et al., ACL 2021) (Hate, Offensive, None)

Davidson et al., ICWSL 2017) (Hate, Offensive, None)

Basile et al., NAACL 2019) (Hate, None), (Aggressive, None)

TASK 2

NLI

Nie et al., ACL 2020) (Entail, Neutral, Contradict) - Facebook

Thorne et al., NAACL 2018) (Entail, Neutral, Contradict) - Wikipedia

TASK 3

Sentiment Classification

Maas et al., ACL 2011) (Neg, Pos) widely known as **IMDB**

Barbieri et al., EMNLP 2020) (Neg, Neutral, Pos)

** Contains various other problem types: hate speech, offensive*

TASK 4

Dialects

Aepli et al., EACL 2023) (en-us, en-gb) + (en) Binary or 3-way

Ziems et al., ACL 2023) 50-label multiclass English dialects

Experiment Conditions
Five-Way Ablation of KD Signals

# / Condition	CE	Logit	Hidden	Attn
① **student_only** *CE only*	ON	OFF	OFF	OFF
② **kd_logit** *CE + KD_logit*	ON	ON	OFF	OFF
③ **kd_logit_hidden** *CE + KD_logit + KD_hidden*	ON	ON	ON	OFF
④ **kd_logit_attn** *CE + KD_logit + KD_attn*	ON	ON	OFF	ON
⑤ **kd_full** *CE + KD_logit + KD_hidden + KD_attn*	ON	ON	ON	ON

Contribution readout

Hidden contribution = ③ − ② · Attention contribution = ④ − ② · Interaction = ⑤ − max(③, ④)

Loss Formulation

Four Loss Terms and Their Combination

Each term captures a different supervision signal in our training framework

TASK
CE Loss

$$L_{\text{CE}} = \text{CE}\left(\mathbf{z}^{(s)},\, y\right)$$

Matches the student to the ground-truth label

LOGIT
Logit KD

$$L_{\text{logit}} = T^2\, D_{\text{KL}}\left(\text{softmax}\left(\frac{\mathbf{z}^{(s)}}{T}\right) \,\Big\|\, \text{softmax}\left(\frac{\mathbf{z}^{(t)}}{T}\right)\right)$$

Matches the teacher's soft predictions with temperature scaling

HIDDEN
Hidden KD

$$L_{\text{hidden}} = \text{MSE}\left(W_h\, \mathbf{h}^{(s)},\, \mathbf{h}^{(t)}\right)$$

Matches intermediate hidden states after projection

ATTENTION
Attention KD

$$L_{\text{attn}} = \text{MSE}\left(A^{(s)},\, A^{(t)}\right)$$

Matches the teacher's attention structure across mapped layers

Total Loss

```
L = α·L_CE  +  β·L_logit  +  γ·L_hidden  +  δ·L_attn
```

each coefficient is 0 or 1 per condition

Preliminary Result

Status & Next Steps — Numbers Coming in the Follow-Up

CURRENT STAGE

Proposal complete · implementation in progress

Preliminary numbers on the four task families will be reported in the next session.

STAGE 1

Teacher fine-tuning

BERT-base on each dataset

in progress

STAGE 2

Baselines

student_only · kd_logit

next

STAGE 3

Multi-level ablations

kd_logit, kd_logit_hidden, kd_logit_attn, kd_full

next

STAGE 4

Analysis & write-up

confusion matrices · contribution · cross-task

final

Deliverable format: the bar-chart comparison (5 conditions × 9 datasets) and confusion-matrix grids will be populated here once runs finish.

Project Proposal

Appendix

Detailed analysis plan

Appendix

Detailed analysis plan - Four Analyses per Experiment

4.1

Overall performance

Accuracy · Macro-F1 · per-class F1 across all 45 runs.

4.2

Confusion matrix

Per-condition confusion matrices — track which class boundaries each KD level resolves, and how confusion patterns shift across tasks.

4.3

Marginal contribution

Hidden = ③ − ② · Attention = ④ − ② · Interaction = ⑤ − max(③, ④).

4.4

Cross-task comparison

Check whether the per-level contribution pattern is consistent across all 9 datasets, both within and across the four task families.